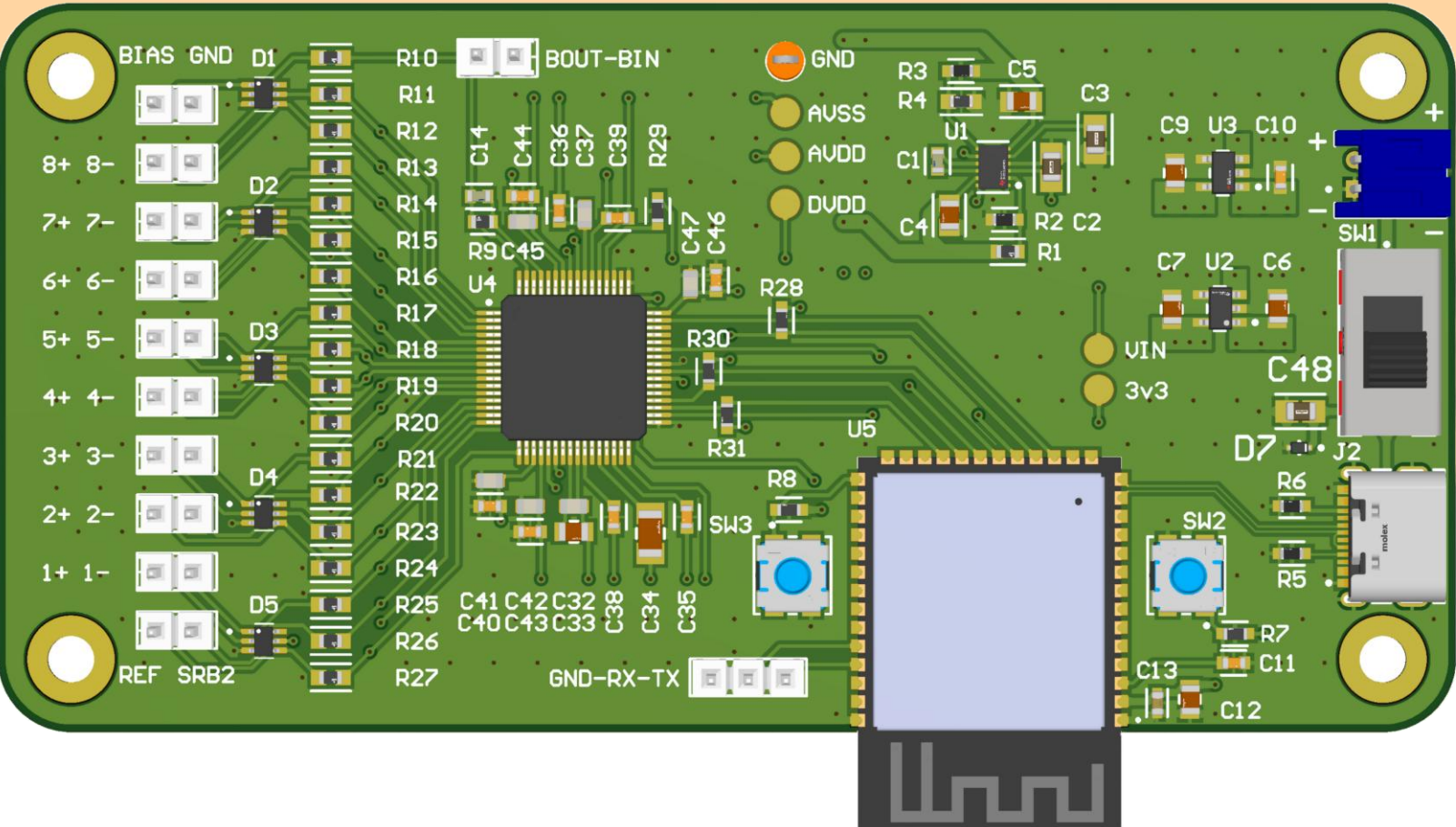
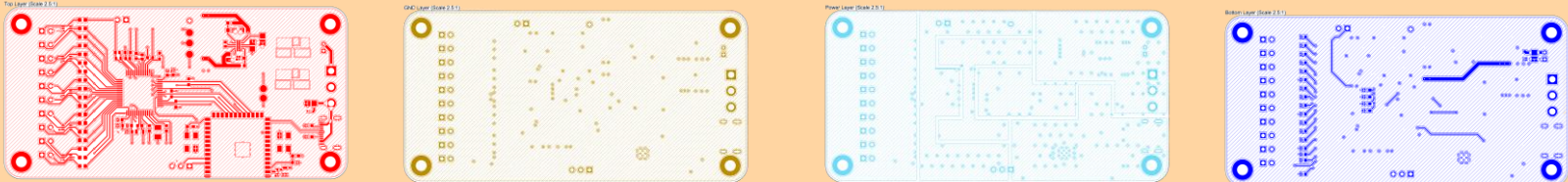
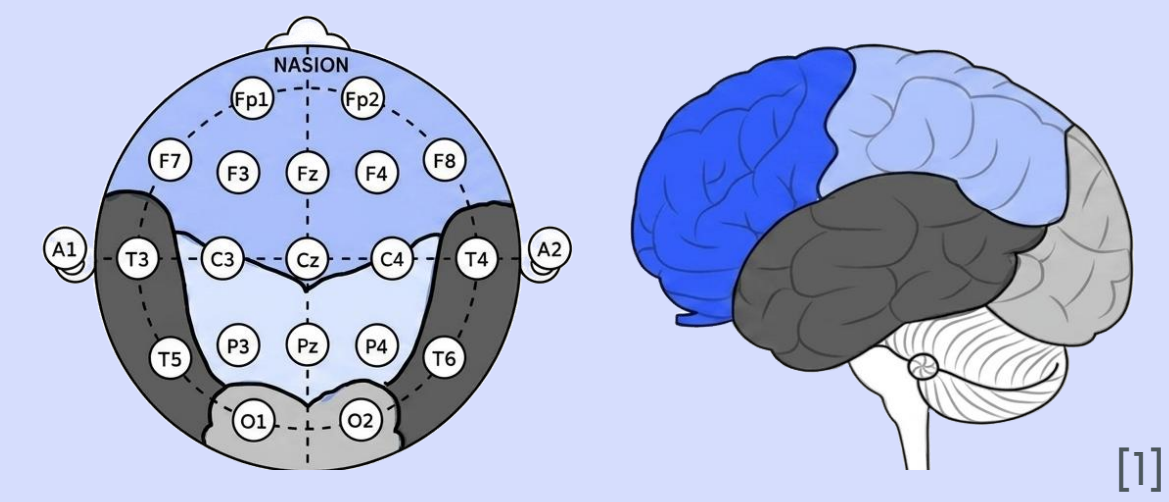


PROTOTYPING A LOW-COST EEG MACHINE

Xavier Asghar – Meng Final Year Project – 2025/26

Introduction

- Electroencephalography (EEG) is a non-invasive method of recording electrical activity in the brain
- EEG has many developing applications, particularly involving **brain-computer interfaces** (BCIs)
- However, commercial EEG headsets are **prohibitively expensive**, costing approximately £1000, while clinical EEG machines cost even more
- This project aims to **lower the cost of EEG technology**, such that it can be utilised in an educational environment, for instance in student research projects, or coursework activities.



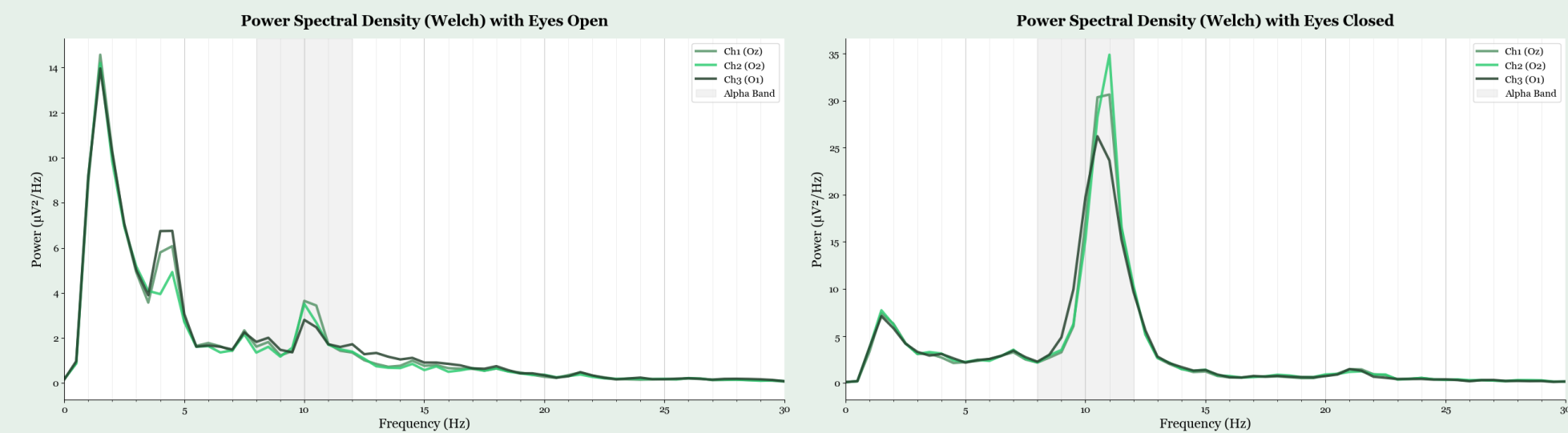
The Hardware

- Up to **8-channels of electrodes** can be connected to the input pin headers
- Texas Instruments ADS1299 EEG Analogue Front-End amplifies and digitises the EEG signal at up to 500 SPS
- The ESP32-S3 then reads the EEG data, and **transmits it to a host computer using Bluetooth Low Energy**
- A python script on the host computer unpacks the EEG data, and pushes it to the Lab-Streaming-Layer, for **visualisation and recording**
- The module is battery powered to ensure patient safety via electrical isolation
- The built in driven right leg helps to cancel common-mode noise

WHAT IT DETECTS

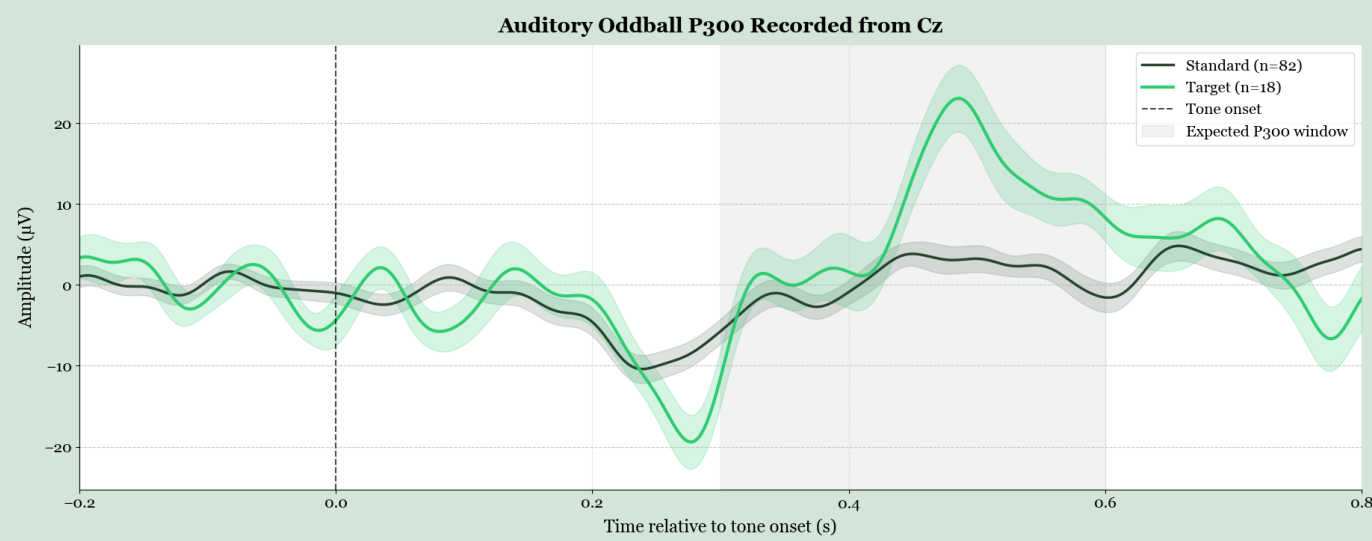
Alpha Waves

- Alpha waves are **8 – 12 Hz oscillations**, recorded from the occipital region of the cortex, from an awake subject with their eyes closed.
- They are thought to represent the functional inhibition of the brain [66]



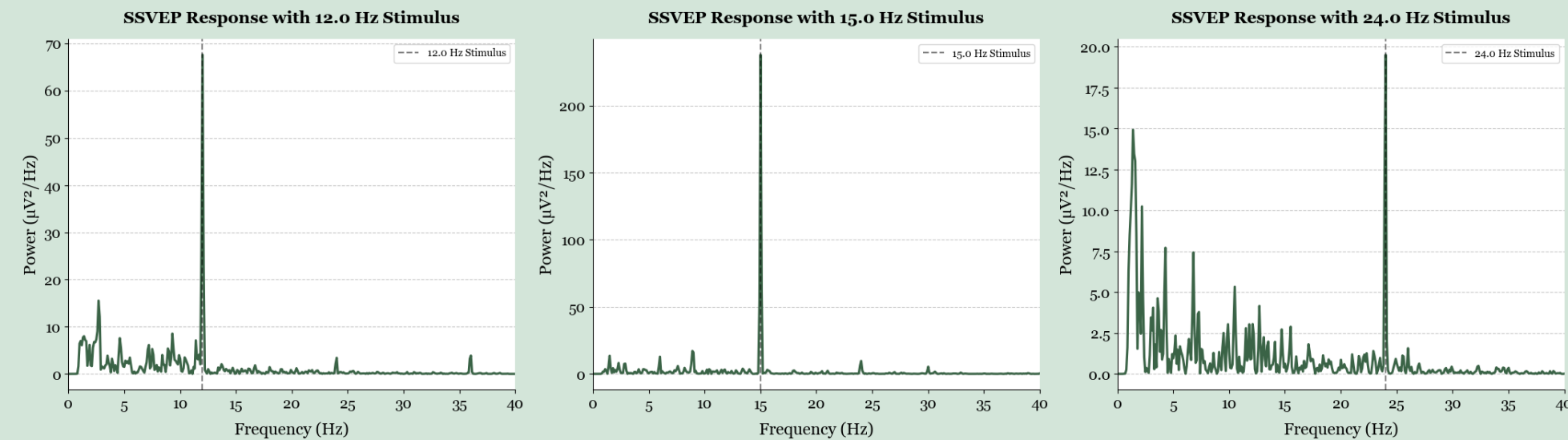
P300 Wave

- The P300 is a positive deflection occurring approximately 300 ms after a subject detects an informative, **task relevant stimulus** [38]
- The exact origins of the P300 are not yet fully understood.



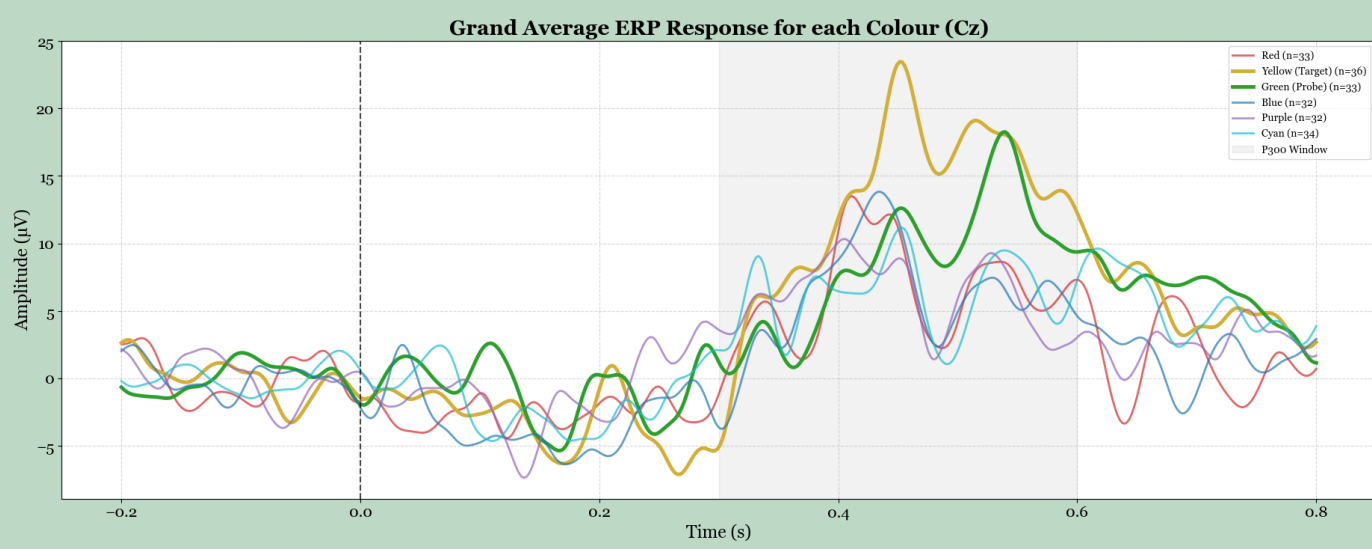
Steady State Visually Evoked Potentials (SSVEPs)

- SSVEPs are evoked responses induced by flickering stimuli, appearing as an **increase in amplitude at the frequency of the stimulus** [70]
- SSVEPs are often used for BCI applications, e.g. BCI spellers



Concealed Information Protocol

- The P300 can be used as a **forensic tool** for memory detection [78]
- The subject chose a secret colour (green), and had to count a target colour (yellow). The response to different colour flashes has been recorded:



CONCLUSIONS

- The proposed system has been proven to accurately measure EEG features commonly used in BCIs
- The module has a minimum **component cost of just £68**
- This represents an almost order of magnitude reduction compared to commercial EEG systems, like the OpenBCI Cyton module (£940), or the Emotiv EPOC X headset (£885)
- With this board it would now be feasible to use EEG technology in a final year project, or give to students as part of a coursework task



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